

# ASSETFLOW – ENTERPRISE IT ASSET MANAGEMENT PLATFORM WITH INTELLIGENT ASSET TRACKING & LIFECYCLE MONITORING

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# ABSTRACT

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AssetFlow is a centralized web-based system designed to simplify and automate IT asset and inventory management. It helps organizations efficiently track devices such as laptops, desktops, monitors, printers, networking equipment, and accessories.

The system allows authorized users to add, update, categorize, assign, transfer, and monitor assets. It maintains details such as asset ID, category, purchase information, employee, location, condition, availability, and status. A dashboard provides insights into total, available, assigned, and maintenance assets.

The frontend uses React.js, while Spring Boot 3 provides RESTful APIs and backend services. PostgreSQL stores asset and user data securely. Docker supports deployment, while GitHub and Jira enable version control, collaboration, and Agile project management.

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# Existing Model

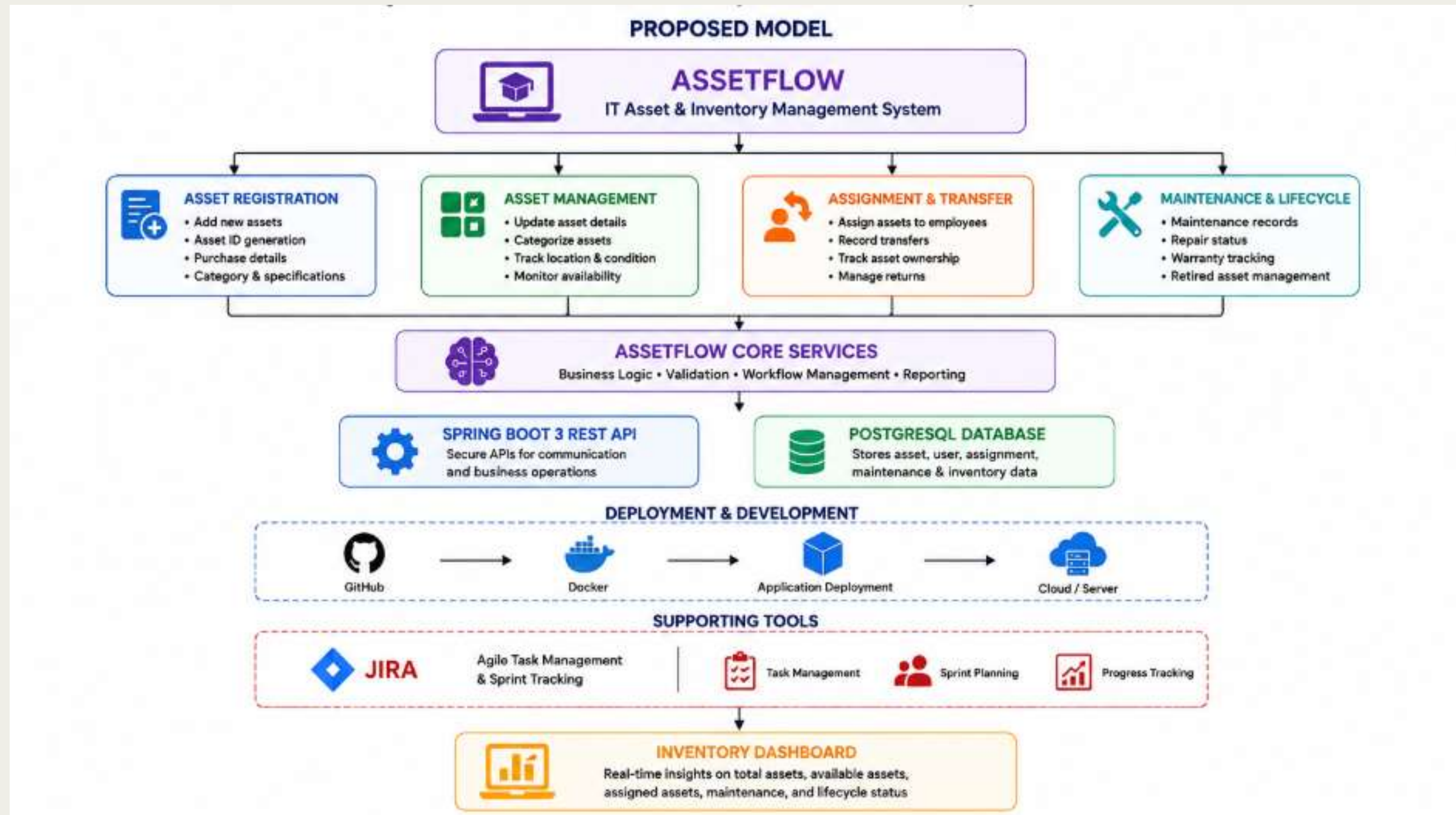
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| Existing System                   | Main Features                                                              | Limitation / Gap                                                                |
|-----------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Snipe-IT</b>                   | Asset tracking, assignments, licenses, locations, and inventory management | Mainly focused on asset tracking with limited workflow automation               |
| <b>ManageEngine AssetExplorer</b> | IT asset tracking, lifecycle management, software and hardware inventory   | Complex for small organizations and requires extensive configuration            |
| <b>ServiceNow ITAM</b>            | Asset lifecycle, service management, procurement, and tracking             | Expensive and more suitable for large enterprises                               |
| <b>Asset Panda</b>                | Asset tracking, barcode scanning, custom workflows, and reporting          | Customization can be complex and costly                                         |
| <b>Spreadsheet-Based Systems</b>  | Basic asset records, employee assignments, and inventory tracking          | Manual updates, duplicate data, data inconsistency, and poor real-time tracking |

# PROPOSED MODEL

AssetFlow integrates asset registration, inventory tracking, assignment, transfer, maintenance, and lifecycle monitoring into a single centralized system.

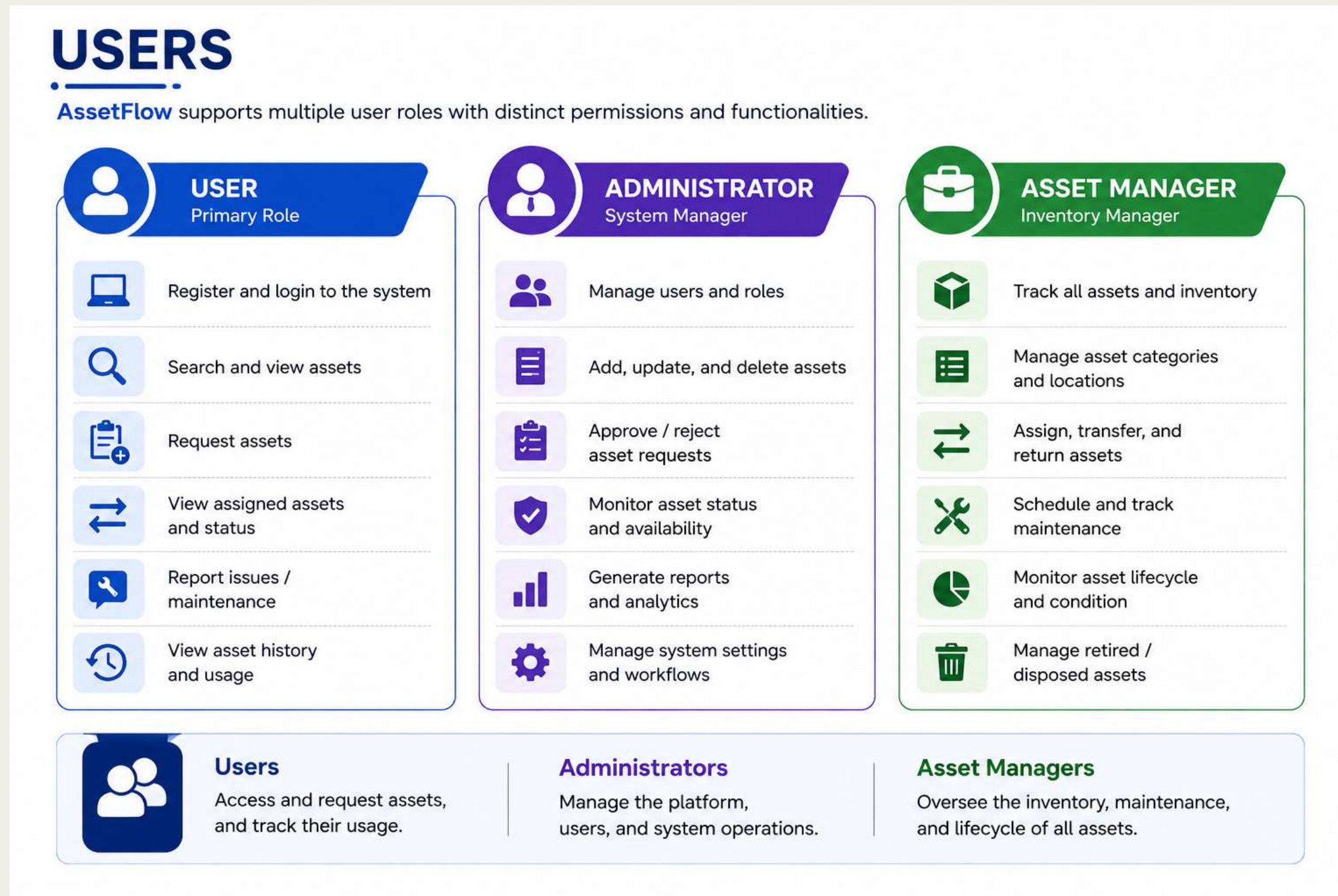
Admin/User → Asset Registration → Asset Categorization → Asset Assignment → Asset Tracking → Maintenance Monitoring → Asset Return/Transfer → Lifecycle Status → Inventory Dashboard



# Users

AssetFlow integrates asset registration, inventory tracking, assignment, transfer, maintenance, and lifecycle monitoring into a single centralized system.

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# Methodology

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Requirement Analysis



Agile / Scrum Planning



System Design & Database Design



Frontend & Backend Development



Asset Management Module Integration



Testing & Validation



Docker Deployment & GitHub Integration



Monitoring & Continuous Improvement

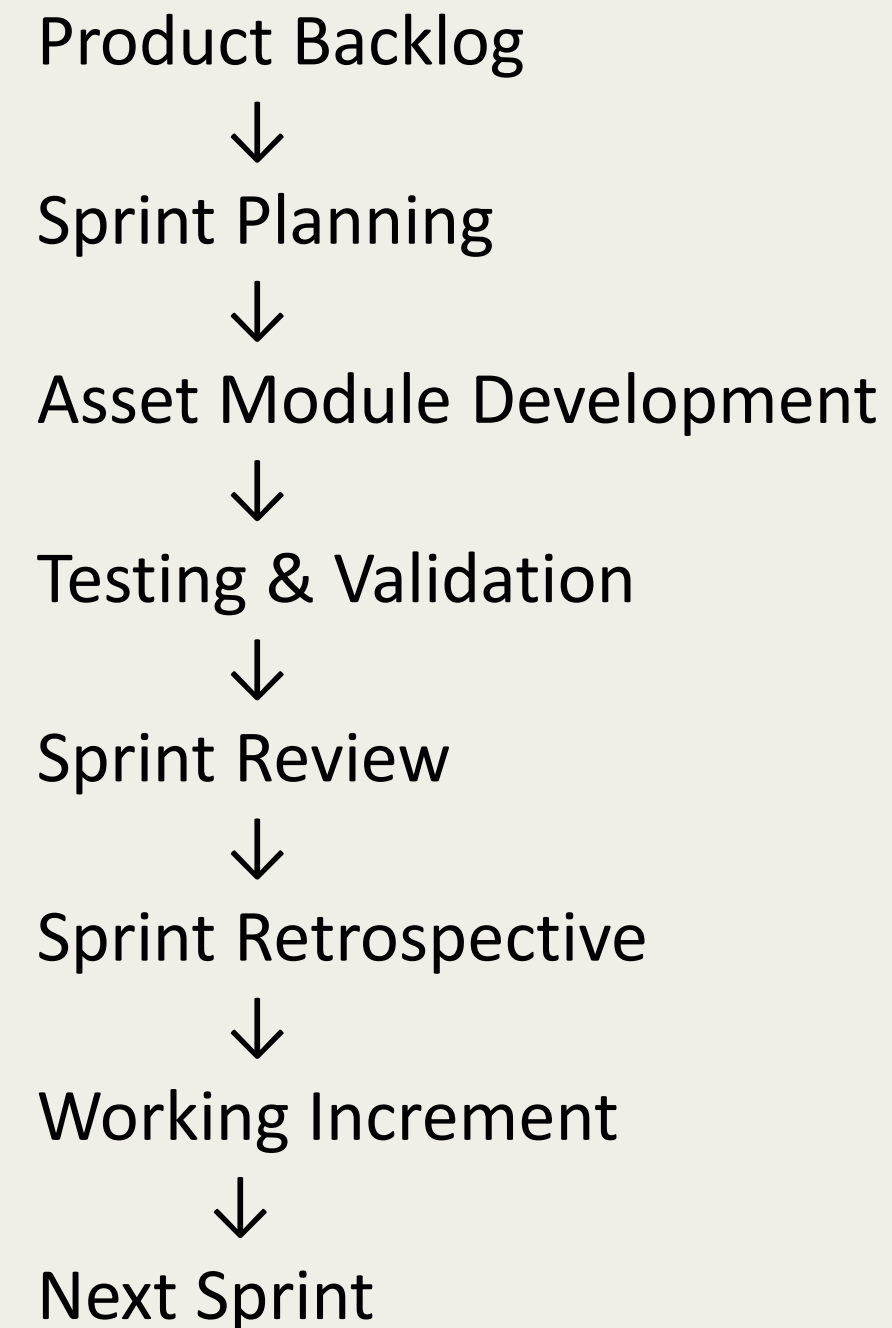
# Process Model

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## Why Agile Scrum?

- **Iterative development** – Build AssetFlow in small functional modules.
- **Continuous feedback** – Get regular feedback from users and team members.
- **Easy adaptation** – Quickly adapt to changing asset-management requirements.
- **Frequent testing** – Test and improve each module after every sprint.
- **Team collaboration** – Supports coordination among developers and stakeholders.

## Agile Scrum Model



# Technical Aspects

## — TECHNICAL ASPECTS —

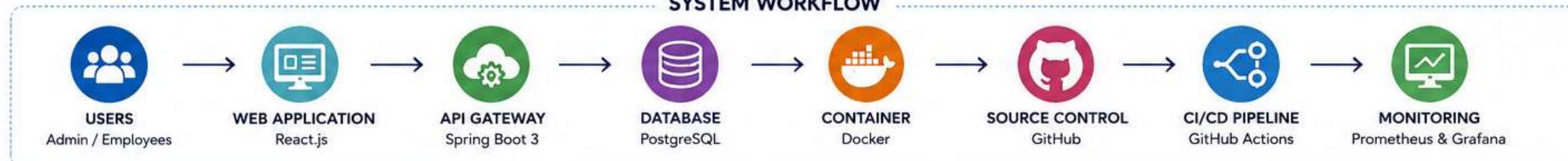
AssetFlow is built using modern technologies and best practices to ensure secure, **scalable and efficient IT Asset & Inventory Management**.



### SUPPORTING PRACTICES



### SYSTEM WORKFLOW



# CONCLUSION

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AssetFlow provides a centralized and efficient solution for managing an organization's IT assets throughout their complete lifecycle. By combining React.js, Spring Boot, PostgreSQL, Docker, GitHub, and Jira, the system enables secure asset tracking, assignment, maintenance, and inventory management.

It reduces manual effort, improves data accuracy, and provides better visibility into asset availability and usage. The modular and scalable architecture also allows AssetFlow to be extended with additional features in the future.

AssetFlow ultimately helps organizations achieve smarter asset utilization, improved accountability, and more efficient IT operations.

THANK YOU